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Follow (or don't follow) the crowd: Young children's conformity is influenced by norm domain and age

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ABSTRACT

This study investigated whether young children's conformity to a consensus varies across the normative domain and age. A total of 168 3- and 5-year-olds participated. Each child was presented with a puzzle box that had two transparent compartments. In a reward preference condition, one of the compartments contained 1 sticker, whereas the other compartment contained 12 stickers. In perceptual judgment and arbitrary preference conditions, one compartment contained a short plank, whereas one contained a perceptually longer plank. Each child was shown a video of four female adults, each of whom was asked the same question within condition: "Which one's the biggest?" (perceptual task; each model retrieved the smaller block), "Which one do you want?" (reward preference; each model retrieved the smaller reward), and "Which one do you want?" (arbitrary preference; each model retrieved the smaller plank). Children were then asked the same question by condition and were allowed to retrieve the item. Notably, more children conformed in the arbitrary preference condition than in the reward preference and perceptual judgment conditions, with 3-year-olds conforming significantly more than 5-year-olds. The 5-year-olds were more successful and imitated with greater fidelity, including demonstrating overimitation. However, less overimitation was observed in the arbitrary preference condition. Together, these findings show that children are sensitive to the contextual cues of the domain in which they

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are witnessing norms and vary their own conformity based on such cues. Furthermore, children can navigate which information to copy to fulfil their own ends.

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Introduction

Conformity to norms has a powerful influence on individual judgments, attitudes, and behavior, as demonstrated by the classic work of Sherif (1935, 1936, 1937) and Asch (1951, 1955, 1956). Due to the critical role of conformity in our choices and behavior, it is essential that we understand how judgments arise in the face of conflicts in normative information; notably, the conflict between individuals' personal information and the majority's behavior (Asch, 1951) as well as between different types of information presented by norms, "injunctive" information about what one should do and "descriptive" information about what the majority does (Deutsch & Gerard, 1955). "Social norms" and "conformity" are terms with many uses, and as a result there is no consensus about meaning (Bicchieri & Muldoon, 2014; Chung & Rimal, 2016). For the purpose of the current study, we use "norm" to mean the behavior adopted by the majority of a group and "conformity" to mean behaving in line with the majority behavior. A "consensus" is an unanimous group behavior. For instance, if most individuals do not litter, this is the norm; if all individuals do not litter, that is a consensus; and if I do not litter because of the influence of the group, I am conforming. Recently, the influence of norms has been examined in the context of children's learning, addressing questions relating to how children use norms to guide their own learning and behavior (e.g., Corriveau, Fusaro, & Harris, 2009; Morgan, Laland, & Harris, 2015; Turner, Nielsen, & Collier-Baker, 2014). The aim of the current study was to examine how levels of children's behavioral conformity to the descriptive norm vary by task domain; we discuss our results in terms of the possible processes that may cause the effects found.

Young children conform to social cues provided by majorities; that is, 3- and 4-year-olds prefer the label given to an ambiguous object by a consensus of three individuals rather than the label given by a single individual (Corriveau et al., 2009), showing adoption of descriptive norms. Furthermore, 4-year-olds adopt the behavior of an informant whose response is supported by a group, through the smiles and head nods of two bystanders, over an informant who receives head shakes and frowns from the two bystanders (Fusaro & Harris, 2008), thereby showing support for children's use of injunctive norms. Turner and colleagues (2014) compared 3-year-olds' use of these different forms of normative behavior, finding that children are influenced by descriptive norms over injunctive norms when the two are misaligned.

Conflicts also occur in relation to copying the majority versus the minority. Conforming to conventions is critical to societal functioning because it allows new members of a group to pick up the social norms quickly without needing to understand the rationale behind them. However, it is also essential that individuals do not blindly conform, adopting the behavior of the majority when in fact this may be disadvantageous to the individual as well as to the group as a whole (Del Vicario et al., 2016), which can result in detrimental "information cascades" (Rieucau & Giraldeau, 2009). Furthermore, as well as not conforming to avoid the adoption of inferior behavior, individuals need to break from the status quo for new advantageous innovations to appear within technological and social practices (Dean, Vale, Laland, Flynn, & Kendal, 2014). Thus, it is critical that we understand how normativity affects learning in children and how biases for majority copying versus personal interests and information are navigated.

Explaining fidelity (copying) versus selectivity (using alternative options) in many realms of child learning remains a challenge. Over and Carpenter (2012) stated that although children can appear credulous, on other occasions they can be discriminating and rational in their learning, whether from several models or one model. For instance, research into the imitation of causally irrelevant actions, known as "overimitation," has shown that children (and adults) copy irrelevant actions under many

conditions, including when they believe that the experiment is over (Lyons, Young, & Keil, 2007), when a reward is at stake (Flynn & Smith, 2012; Lyons, Damrosch, Lin, Macris, & Keil, 2011), and when the actions are presented by an individual they believe to be a fellow participant (Flynn & Smith, 2012). On other occasions, children rationally mediate their imitation (Gergely, Bekkering, & Király, 2002; Meltzoff, 1995). Over and Carpenter (2012) argued that three factors moderate fidelity versus selectivity: (a) desire to affiliate with the social group or model, (b) social pressure felt in the situation, and (c) the child's own goals. Others highlight that children construe the situation as being a "ritual" in which behavior is adopted by a conventional function rather than an instrumental function, suggesting that this may explain the discrepancies (Kapitány & Nielsen, 2015; Legare, Wen, Herrmann, & Whitehouse, 2015). Alternatively, Walker and Andrade (1996) provided evidence that situational ambiguity is an important factor for children aged from 3 to 17 years, with greater uncertainty causing more conformity.

Although there is still much research needed to address these differing explanations, several recent studies have cast light on the fact that the domain within which the normative behavior is presented, as well the observing children's age, has an influence on conformity. Seston Schillaci and Kelemen (2014) found that 3- and 4-year-olds deferred to a majority's behavior with regard to object functions. Children were more likely to agree with the majority when majority and minority opinions were equally plausible, especially when the majority demonstrated an overt consensus. However, 4-year-olds actively eschewed the majority opinion when it was implausible in the context of the artifact's functional design; in such cases, 4-year-olds trusted their own judgment over that of the majority. Similarly, Corriveau and Harris (2010) showed that 3- and 4-year-olds deferred to a majority less often when their judgment would be functionally tested than when it was a perceptual judgment. That is, when they were making a judgment about the length of comparative lines, they deferred more than when those comparative lines would be used to build a bridge for a toy protagonist to cross than when they simply needed to state which was longer. Recently, Bernard, Harris, Terrier, and Clément (2015) gave a further demonstration of preschoolers' weighing up of personal versus norm-based information. They found that 3- to 5-year-olds were more likely to rely on social information if personal information was ambiguous and if there was a consensus of three individuals providing information rather than one individual. Furthermore, it was found that 5-year-olds were more likely to rely on personal information than on social information. Normative and moral social factors have also been shown to be important in young children's conformity and imitation (Kim, Chen, Smetana, & Greenberger, 2016; Rakoczy, Warneken, & Tomasello, 2009).

In the current study, we directly tested 3- and 5-year-old children's conformity to a majority's judgment across three normative domains: (a) reward preference, in which the majority selected 1 sticker over 12 stickers; (b) perceptual judgment, in which the majority was presented with two wooden blocks and selected an obviously smaller block when asked which is biggest; and (c) arbitrary preference, in which the majority selected one of two blocks when asked which one they preferred (which clearly had no obviously "correct" answer). We define "normative domains" as contexts that draw on different cognitive and social processes. This study presents two critical extensions to the current understanding of conformity. First, it addresses whether young children show differing levels of conformity across different norm domains using a standardized procedure across each of these domains. In line with previous research, we predicted less conformity in norm domains in which there is a clear contrast with one's own perception or desires (the reward preference and perceptual judgment domains) compared with domains in which the rationale for the majority's judgment is not clear or may indeed be conventional (the arbitrary preference domain) and there is no clear contrast with one's own judgment. Such domain differences may also be influenced by the age of the participants, with younger children (3-year-olds) being more likely to conform across all the domains than older children (5-year-olds), mirroring developmental changes seen in other studies (Corriveau & Harris, 2010; Seston Schillaci & Kelemen, 2014; Walker & Andrade, 1996). The age groups chosen represent the beginning of early social development and, at 5 years, move into middle childhood. Thereby, they give us a clear picture of changes in norm use in early development, fitting with the majority of previous research in this area, which focuses on 3- to 5-year-olds (Corriveau & Harris, 2010; Seston Schillaci & Kelemen, 2014; Walker & Andrade, 1996).

Recently, there has been increased interest in how aspects of context, including the social context, affect copying behavior and the implications this might have for learning and cultural acquisition, transmission, and evolution more broadly (Reader, Morand-Ferron, & Flynn, 2016). The current study adds to this literature by exploring whether a majority's behavior, which was clearly inaccurate (as in the perceptual judgment task), less advantageous (as in the reward preference), or ambiguous (as in the arbitrary preference), influences a young child's subsequent behavior in terms of the level of fidelity the child demonstrates to the behavior that he or she witnessed the majority undertake on a task. It could be argued that when majorities appear to be less accurate or undertake less advantageous behavior, they are less likely to be copied across other behaviors than majorities whose behavior is ambiguous. For example, when a majority's behavior was pitted against success, such that the majority's actions were unsuccessful in opening a puzzle box while the minority's behavior was successful, 4- and 5-year-olds copied the behavior of the minority (Wils, Collier-Baker, & Nielsen, 2015). Thus, we predicted that 3-year-olds should show greater fidelity, in terms of action replication including overimitation, when the majority's selection of an object is ambiguous than when it contrasts with a child's own preferences or perceptions compared with 5-year-olds. "Overimitation" refers to children's proclivity to copy obviously causally redundant actions and has been argued to be influenced by the same factors as imitation and conformity (see Kenward, Karlsson, & Persson, 2010; Keupp, Behne, & Rakoczy, 2013; Lyons et al., 2007; Nielsen & Blank 2011). That is, although both judgment and imitation have been argued to be influenced by similar sets of social influence (Over & Carpenter, 2012), our study may give further information about how these phenomena are related. Addressing such questions establishes whether children are sensitive to the contextual cues of the domain in which they are witnessing the conformity of others' behavior, varying their own conformity and subsequent behavior based on such cues.

Method

Participants

Participants were 168 children from schools and nurseries in North East England. They were drawn from two age groups: 3-year-olds ($n = 84$, 40 girls; $M = 44.65$ months, $SD = 3.50$) and 5-year-olds ($n = 84$, 38 girls; $M = 67.68$ months, $SD = 3.35$). The majority of children were White British, with Asian being the second most represented ethnic group. Informed consent was provided by the children's parents and the nursery or school staff. In addition, all children verbally consented to participate when asked whether they wished to take part. Ethical approval was given by the School of Education's ethics committee at Durham University.

Design

A 2×3 between-groups design was used to assess whether age (3-year-olds or 5-year-olds) and norm domain (reward preference, perceptual judgment, or arbitrary preference) influenced conformity. The key outcome variable was whether a child's choice of an object matched the choice of the group (conformed) or the child selected an alternative option (did not conform). We also investigated, across these different groups (age and norm domain) whether children imitated the same sequence of actions as the group to remove the selected object from a puzzle box (imitation fidelity) or whether they did not replicate faithfully; this imitation fidelity included a measure of overimitation demonstrated by the majority. The children's success on the task, acquisition of the object from a puzzle box, was also recorded.

Apparatus

The Duobox was the apparatus used in this study (see Fig. 1). It has two conjoined compartments which are identical apart from the color painted on the back wall of the apparatus; one compartment was blue, and one was red. The apparatus was transparent, such that the object contained inside each

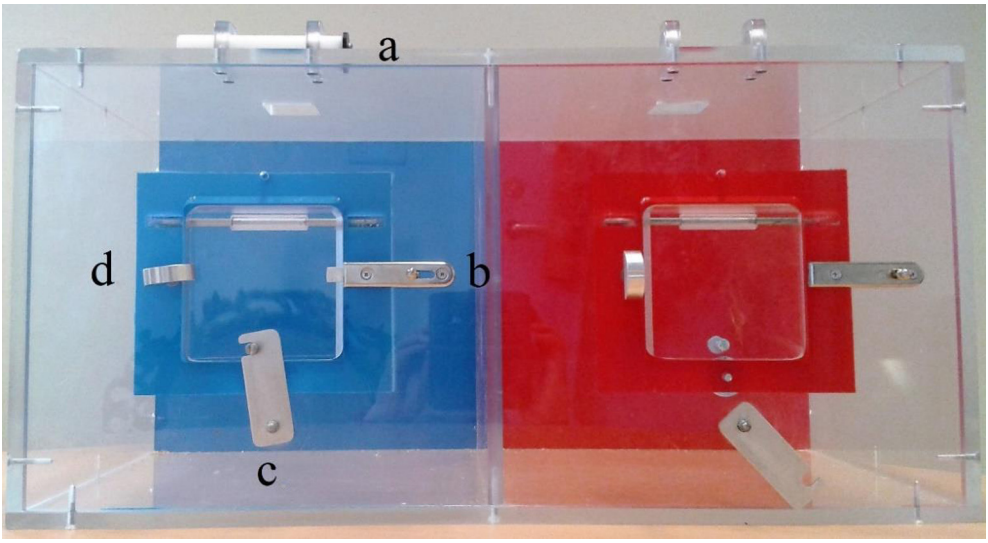

Fig. 1. The Duobox. The left side shows the box in assembled state. The right side shows the box with defenses removed: (a) bolt; (b) lock; (c) hook; (d) latch.

compartment could be seen easily. The sequence of actions modeled was as follows. First, the bolt was removed from the top of the apparatus (labeled "a" in Fig. 1), a causally irrelevant action allowing the study of overimitation. Then three different defenses were removed to allow a door to be opened and the object retrieved; that is, a horizontal latch was twisted to a vertical position (labeled "b" in Fig. 1), a hook was pulled clockwise (labeled "c" in Fig. 1), and a flat bolt was pulled to the right (labeled "d" in Fig. 1). In the reward preference condition, one of the compartments contained 1 sticker, whereas the other one contained 12 stickers identical in shape and size but with varying designs, including three replicates of the alternative single sticker; fully 100% of both 3- and 5-year-olds preferred the 12 stickers to the 1 sticker in pilot testing ($N = 20$). In the perceptual judgment and arbitrary preference conditions, one compartment contained a short plank of wood, whereas the other one contained a long plank. The long plank was three times the length, and pilot testing found that 100% of both 3- and 5-year-olds correctly identified their differential length from their position in the apparatus ($N = 20$).

Conformity stimulus

To establish the descriptive norm, children were shown a video of four female adult models opening the Duobox. Initially the four models stood side by side, and then each model stepped forward in turn and opened the box using identical actions. Thus, children saw each model (a) step forward from the group, (b) be asked a question regarding her perception or preference (differing based on condition), (c) perform the same action sequence on the Duobox, (d) retrieve the reward, and (e) return to the group. After each model retrieved the item from the Duobox, she held it toward the camera and smiled.

In the reward preference condition, each model's actions were preceded by the question "Which one do you want?", with all models retrieving the small reward. In the perceptual judgment condition, each model's actions were preceded by the question "Which one's the biggest"?, with all models retrieving the smaller plank despite it being perceptually smaller. In the arbitrary preference condition, each model's actions were preceded by "Which one do you want?" All models retrieved the smaller plank; there was no obvious natural preference for either plank. The side, red or blue, was consistent within the video with the models and within each child's attempt but was counterbalanced across participants within conditions.

Procedure

Testing took place in a quiet room away from other children within children's school or nursery. After a short settling period, children were shown the apparatus and asked a series of questions to clarify that they understood that (a) it had two sides, (b) the sides were different colors, and (c) the contents of each side were different. Children then watched one of the conformity stimuli videos. They were told that their job was to get something from inside the puzzle box. Following this, children were asked the same questions as those asked in the video ("Which one do you want?" in the reward preference and arbitrary preference conditions and "Which one's the biggest?" in the perceptual judgment condition). They were then allowed to retrieve the object from the Duobox. All children were thanked for their time and received a sticker reward.

Coding

For the conformity measure, the children's actions were coded as either matching the group (coded 1) or selecting the alternative (coded 0) depending on which compartment they interacted with first (this did not differ from the object selected). For imitation fidelity, they were coded as either copying the removal of the defenses exactly (coded 1), or using another sequence (coded 0). This was operationalized as successfully unlatching each defense in the exact order that had been demonstrated. This dichotomous coding was found to be the most explanatory way to code fidelity in this context. For overimitation, children were coded as having overimitated if they removed the causally irrelevant bolt (coded 1) or as not overimitating if they did not (coded 0). Finally, children were coded as successful if they retrieved the object from within the Duobox within 5 min (coded 1) or as unsuccessful if they did not meet this criterion (coded 0). Testing was discretely recorded and was performed on the recorded data rather than live.

Contrast coding was employed to reflect the following key theoretical comparisons. The first was between the arbitrary preference (coded 2) and conditions in which there was a motivation to depart from the observed consensus, the reward preference (coded −1) and perceptual judgment (coded −1) conditions. The second comparison was between the two competing motivation conditions: reward preference, coded 1, and perceptual judgment, coded −1, with arbitrary preference coded 0. For age, the 5-year-old group was coded 1 and the 3-year-old group was coded −1. When conformity was entered as a predictor, conforming was coded 1 and not conforming was coded −1.

Results

To examine how age and norm domain affected young children's conformity, imitation fidelity, overimitation, and success, binary hierarchical logistic regressions were performed (bootstrapping 10,000 iterations). At Step 1, age group and norm domain were entered as predictors. For analyses of performance measures (imitation fidelity, overimitation, and success), conformity was also entered at Step 1. The interactions between age group and norm domain were entered at Step 2. Predictor analyses are reported in Table 1, and descriptive norm domain by age group cell descriptive statistics are reported in Table 2. Our statistical model is performed in two steps to reflect our prioritization of main effects and then interactions. This statistical approach allowed us to examine differences based on experimental condition (which was of most theoretical importance given our interest in the effect of normative domain), age, and then how these factors interacted. Where R^2 is reported, it is the Nagelkerke R^2.

Conformity

At Step 1, the model was significant, $R^2 = .14$, $\chi^2(3, N = 168) = 17.93$, $p < .001$. Children demonstrated higher conformity in the arbitrary preference condition than in the reward preference and perceptual judgment conditions odds ratio (OR) = 1.44, 95% confidence interval (CI) [1.15, 1.87]. However, there was no significant difference in conformity between the reward preference and perceptual

Table 1
Hierarchical binary multiple regression analyses of age and norm domain on outcome variables (conformity, imitation fidelity, overimitation, and success).

	Conformity			Imitation fidelity			Overimitation			Success		
	β	SE	p	β	SE	p	β	SE	p	β	SE	p
Step 1												
Intercept	−.43	.17	.010*	−1.39	.28	.001*	−.11	.18	.529	2.28	2.11	.001*
Age	−.43	.18	.011*	.49	.22	.014*	−.60	.17	.001*	1.01	2.00	.002*
Arbitrary preference vs. reward preference & perceptual judgment	.36	.12	.002*	−.24	.21	.128	−.26	.13	.032*	−.13	.32	.469
Perceptual judgment vs. reward preference	.26	.22	.212	.17	.23	.434	−.25	.21	.210	−.24	.81	.467
Conformity				−.28	.24	.205	−.05	.18	.788	−.54	.46	.034*
Step 2												
Intercept	−.46	.50	.006*	−1.38	.88	.001*	−.12	.24	.503	2.25	.33	.001*
Age	−.46	.50	.006*	.43	.88	.020*	.60	.23	.001*	.95	.33	.004*
Arbitrary preference vs. reward preference & perceptual judgment	.37	.27	.001*	−.20	.69	.172	−.26	.20	.030*	−.02	.23	.919
Perceptual judgment vs. reward preference	.28	.74	.190	.14	.96	.536	−.24	.22	.233	−.11	.40	.783
Age by arbitrary preference vs. reward preference & perceptual judgment	.16	.27	.179	−.23	.68	.108	.03	.20	.840	.16	.23	.482
Age by perceptual judgment vs. reward preference	.06	.73	.758	.11	.95	.612	.25	.21	.210	.19	.40	.638
Conformity				−.27	.25	.251	−.05	.19	.789	−.56	.26	.029*

Note. $N = 168$. Rows contain predictors at each step, and columns contain dependent variables and associated statistics.
* $p < .05$.

Table 2
Descriptive statistics by age group and norm domain condition.

	Perceptual judgment		Reward preference		Perceptual judgment & reward preference		Arbitrary preference	
	%	(*SD*, %)	%	(*SD*, %)	%	(*SD*, %)	%	(*SD*, %)
Conformity	25	(44)	38	(49)	32	(47)	57	(50)
3-year-olds	39	(50)	50	(51)	44	(50)	61	(50)
5-year-olds	14	(36)	25	(44)	20	(40)	54	(51)
Imitation fidelity	25	(44)	30	(46)	28	(45)	14	(35)
3-year-olds	14	(35)	14	(36)	14	(36)	14	(36)
5-year-olds	36	(49)	46	(51)	41	(50)	14	(36)
Overimitation	60	(49)	48	(50)	54	(50)	36	(48)
3-year-olds	52	(51)	29	(46)	40	(49)	21	(42)
5-year-olds	68	(48)	68	(48)	68	(47)	50	(51)
Task success	93	(26)	86	(35)	89	(31)	82	(39)
3-year-olds	89	(32)	75	(44)	82	(39)	68	(48)
5-year-olds	96	(19)	96	(19)	96	(19)	96	(19)

judgment conditions, OR = 1.29, 95% CI [0.87, 2.02]. Furthermore, 3-year-olds (50% conformed, *SD* = 50%) demonstrated significantly higher conformity than 5-year-olds (31% conformed, *SD* = 47%), OR = 0.65, 95% CI [0.45, 0.90].

At Step 2, with the interaction terms in the model (for depiction, see Fig. 2), step change was non-significant, $\Delta R^2 = .01$, $\chi^2(2, N = 168) = 1.84$, $p = .398$, but the full model was significant, $R^2 = .15$, $\chi^2(2, N = 168) = 19.78$, $p = .001$. The same pattern of results was observed for age group, OR = 0.63, 95% CI [0.37, 0.88]. This was also the case for arbitrary preference versus reward preference and perceptual

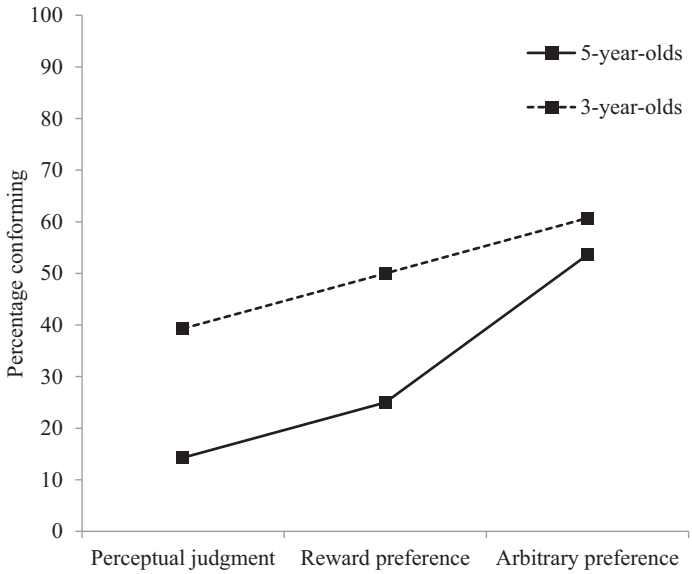

Fig. 2. Proportions of children matching the descriptive norm (conforming) by domain and age group.

judgment conditions comparison, OR = 1.45, 95% CI [1.16, 2.12], although there was no significant interaction with age, OR = 1.17, 95% CI [0.92, 1.66]. Again, as in Step 1, there was no significant difference between the reward preference and perceptual judgment conditions, OR = 1.33, 95% CI [0.85, 2.50]; it was also found that there was no interaction with age, OR = 1.07, 95% CI [0.67, 1.96].

Performance measures

When examining imitation fidelity, the model was significant, $R^2 = .12$, $\chi^2(4, N = 168) = 14.02$, $p = .007$. The 5-year-olds copied with higher fidelity than the 3-year-olds, OR = 1.62, 95% CI [1.11, 2.61]. There was no difference in imitation fidelity between arbitrary preference and the reward preference and perceptual judgment conditions, OR = 0.79, 95% CI [0.53, 1.07]. Furthermore, there was no difference in imitation fidelity between those who conformed (15% exact copy, *SD* = 36%) and those who produced the alternative judgment/preference (29% exact copy, *SD* = 46%), OR = 0.75, 95% CI [0.44, 1.15]. With the age group by norm domain interaction terms in the model, the change statistic was nonsignificant, $\Delta R^2 = .02$, $\chi^2(2, N = 168) = 2.48$, $p = .248$, but the full model was significant at Step 2, $R^2 = .14$, $\chi^2(6, N = 168) = 16.51$, $p = .011$. Age group remained significant in the same direction, OR = 1.54, 95% CI [0.97, 40.37]. The arbitrary preference versus reward preference and perceptual judgment conditions comparison remained nonsignificant, OR = 0.82, 95% CI [0.04, 3.73], as did the comparison of the reward preference and perceptual judgment conditions, OR = 1.15, 95% CI [0.63, 2.25]; there was found to be no interaction with age for either of these comparisons, OR = 0.80, 95% CI [0.15, 1.19] and OR = 1.12, 95% CI [0.58, 2.08], respectively.

The model predicting overimitation was significant at Step 1, $R^2 = .16$, $\chi^2(4, N = 168) = 20.60$, $p = .001$. The 5-year-olds (62% overimitated, *SD* = 49%) overimitated significantly more than the 3-year-olds (34% overimitated, *SD* = 48%), OR = 1.81, 95% CI [1.34, 2.61]. Children in the arbitrary preference condition overimitated less when compared with the reward preference and perceptual judgment conditions, OR = 0.77, 95% CI [0.59, 0.98]. However, there was no difference in performance of overimitation behavior between the reward preference and perceptual judgment conditions, OR = 0.78, 95% CI [0.51, 1.19]. There was also no difference in overimitation between those children who conformed (40% overimitated, *SD* = 49%) and those who did not (53% overimitated, *SD* = 50%), OR = 0.95, 95% CI [0.66, 1.33]. At Step 2, the change statistic was nonsignificant, $\Delta R^2 = .01$,

$\chi^2(2, N = 168) = 1.56$, p = .459; the full model was significant, $R^2 = .17$, X^2 (6, N = 168) = 22.16, $p = .001$. The pattern of results was not changed with the interaction terms in the model. Age was a significant predictor, OR = 1.82, 95% CI [1.32, 2.75]. There was a significant difference between arbitrary preference and the reward preference and perceptual judgment conditions, OR = 0.77, 95% CI [0.57, 0.98], although the interaction with age was nonsignificant, OR = 1.03, 95% CI [0.80, 1.37]. The difference between the reward preference and perceptual judgment conditions was nonsignificant, OR = 0.79, 95% CI [0.51, 1.21], as was its interaction with age, OR = 1.28, 95% CI [0.85, 1.98].

In terms of task success, the model was significant at Step 1, $R^2 = .23$, $\chi^2(4, N = 168) = 23.01$, $p = .001$. The 5-year-olds (96% successful, $SD = 19\%$) were significantly more successful than the 3-year-olds (76% successful, $SD = 43\%$), OR = 2.74, 95% CI [1.55, 21741.97]. There was no difference in level of success between arbitrary preference and the reward preference and perceptual judgment conditions, OR = 0.88, 95% CI [0.58, 1.32]. There was also no difference between reward preference and perceptual judgment, OR = 0.79, 95% CI [0.34, 1.55]. In terms of conformity, those who conformed were less successful (77% successful, $M = .76$, $SD = .43$) than those who did not (93% successful, $M = .93$, $SD = .26$), OR = 0.58, 95% CI [0.28, 0.98]. At Step 2, the change statistic was not significant, $\Delta R^2 = .01$, $\chi^2(2, N = 168) = 0.68$, $p = .712$; the full model was significant, $R^2 = .24$, $\chi^2(6, N = 168) = 23.69$, $p = .001$. The pattern of results was not changed with the interaction terms in the model. The 5-year-olds were more successful than the 3-year-olds, OR = 2.59, 95% CI [1.49, 20050.31]. There was no difference in success in arbitrary preference as compared with the reward preference and perceptual judgment conditions, OR = 0.98, 95% CI [0.04, 23.10], and the interaction of this comparison with age was also nonsignificant, OR = 1.18, 95% CI [0.05, 1.32]. Likewise, there was no difference between reward preference and perceptual judgment, OR = 0.90, 95% CI [<0.01, 112.17], or their interaction, OR = 1.21, 95% CI [0.01, 165.17]. However, again, children who conformed less were more successful, OR = 0.57, 95% CI [0.27, 0.98].

Discussion

We found that young children were more likely to conform to a consensus in the domain of an arbitrary preference condition compared with when there was a larger reward at stake or a correct perceptual judgment to make. The 3-year-olds were more likely to conform than the 5-year-olds. When considering the replication of the specific actions on the task, 5-year-olds were more faithful to the observed actions than 3-year-olds, and this included overimitating. Interestingly, children showed more overimitation in the reward preference and perceptual judgment conditions than in the arbitrary preference condition. The 5-year-olds were also more successful compared with the 3-year-olds irrespective of the condition, with those children who did not conform being more successful at the task.

A pivotal challenge for understanding children's learning and innovation is to establish when and why selectivity overrides fidelity (Carr, Kendal, & Flynn, 2015, 2016; Over & Carpenter, 2012). The current study found that with regard to conformity, both domain and age influenced children's willingness to copy the actions of a unanimous majority. That is, rather than conforming blindly, children process the domain-relevant information and modulate their conformity in response to it. Children conformed more to demonstrated consensus when the norm demonstrated was arbitrary rather than in a domain marked by a conflict with personal information, as in the perceptual judgment task, or against personal interest, as in the reward preference task.

At a gross level, this finding shows that when there is a competing behavioral tendency (such as a desire to be correct or for a large reward), conformity will diminish. Theorists give us several potential explanations for this effect. First, arbitrariness is a cause of ambiguity, which may result in a child not knowing what underpins the cause of a preference (in the case of the arbitrary preference condition, there is no cause) and, therefore, adopting a strategy of conforming (Walker & Andrade, 1996). Such a bias has been called a "copy-when-uncertain" strategy (Laland, 2004), with the logic being that when one is uncertain, taking on the group behavior is likely to lead to a more advantageous strategy than adopting a chance behavior. Second, it may be that the child perceives the observed choice as purely conventional, or part of ritual, and thereby copies for injunctive/normative reasons; that is, the

reproduction of the actions is the culturally right thing to do in this context (Kapitány & Nielsen, 2015; Legare et al., 2015). Such normative behavior allows affiliation and smooth integration with group members, as well as the acquisition of instrumental information, even when the causes of the behavior are opaque (Over & Carpenter, 2012). Teasing apart these normative versus informational motivations for conformity is a difficult undertaking for researchers. For example, it is notable that within our experiment the perceptual judgment task contains a competing social goal given that the children needed to announce their response to the experimenter, and socially relevant factors such as presenting a response in public versus private have been shown to modulate children's conformity (Haun & Tomasello, 2011). But it appears that despite such social pressure to copy the majority, under some contexts, including when selecting which of two blocks is longer, the normative pressure to be correct is more powerful than the bias to copy a majority. In terms of the absolute levels of conformity observed, they were highest in the arbitrary preference condition (57%). Corriveau and Harris (2010) consistently observed conformity levels under 50%, whereas others have found higher levels (Bernard et al., 2015). The current experiment attests to the possibility that between-task differences are likely to be important. Furthermore, it is known that the culture of the children in an experiment has a substantial impact (Corriveau & Harris, 2010; Corriveau, Kim, Song, & Harris, 2013). Similar previous research has shown that between-condition effects are consistent over trials, with conformity potentially diminishing (Bernard et al., 2015; Corriveau & Harris, 2010). This suggests that the pattern of results in the current experiment, using a single trial, is valid, although future research is needed to properly address how conformity levels change over time and experience.

The current study corroborated previous research findings regarding children's greater willingness to eschew conformity as they develop (Corriveau & Harris, 2010; Seston Schillaci & Kelemen, 2014; Walker & Andrade, 1996). Such results accord well with Flynn, Turner, and Giraldeau (2016), which suggests that 5-year-olds show more selectivity and understanding of when to deploy social (as well as asocial) information than 3-year-olds. Although not borne out by inferential statistics in the current study, 5-year-olds may have been tending toward similar levels of conformity to 3-year-olds in the arbitrary condition (4% difference) but showed an apparent proclivity to not conform when there was a compelling competing reason (25% difference in both conditions). Future research may see whether such a difference is reliable if an older cohort of children is considered.

Likewise, our results reflect previous research showing that the quality and accuracy of imitation increase with age (McGuigan, Makinson, & Whiten, 2011; McGuigan, Whiten, Flynn, & Horner, 2007). In the current study, this was evident in increased fidelity, overimitation, and success in 5-year-olds compared with 3-year-olds. Together, these age results suggest that children become more competent, over development, at undertaking the learned behavior or norms to which they have been exposed. Furthermore, it was found that conforming children were less successful—that is, less able to complete the task of retrieving the object from the apparatus—than nonconforming children. This makes sense in the light of Flynn and colleagues (2016), who found that children (5-year-olds specifically) who chose to learn individually, as opposed to socially, were more adept at completing a novel apparatus or tool-use task than children who wished to learn socially but received asocial learning instead. A potential explanation here is that children who depart from socially structured behavior do so because they have succeeded in using individual learning before or can deduce an efficient solution.

By comparing normative domain judgment and imitation behavior findings, the current study again highlights children's growing appreciation of how best to implement available information in their own behavior, especially by 5 years of age (in line with Flynn et al., 2016). Children are willing to eschew normative pressure to fulfill some motivated goal, as in attaining a reward, but may retain the methods used to achieve a goal. This suggests a hypothesis that children become more adept, by the end of early development, at deploying available social information from the same sources about different aspects of a task (goal vs. methods) to meet their own ends. Furthermore, departing from demonstrated goals and methods is bounded with children's capacity to efficaciously acquire success with their own techniques (Flynn et al., 2016).

It is notable that children overimitated more in the reward preference and perceptual judgment conditions than in the arbitrary preference condition. It would seem hard to reconcile this difference with the hypothesis that overimitation in this study was the result of distorted causal understanding

(e.g., Lyons et al., 2007, 2011); that is, children were more confused about whether the redundant actions were efficacious or not in the conditions with a competing motivation than when the choice was arbitrary (especially given the ability of children to dissociate goals and actions described above). Rather, our conjecture would be that the social expectations of the context may be playing a role and that the overimitated action can be seen as part of an injunctive norm—a thing one is supposed to do (Kenward et al., 2010; Keupp et al., 2013; Nielsen & Blank, 2011; Over & Carpenter, 2012). Specifically, in a context in which there was no right answer, there were no "correct" actions to perform, whereas, when there is a correct or better option, a proportion of children may have inferred that there was a correct way to perform the task, including the redundant action (which in reality posed little cost in terms of time).

Our results provide important insights into the development of conformity showing when, and under which conditions, young children copy the majority versus when they undertake an alternative. For society to function effectively in terms of both technological and social systems, we must conform to a consensus. However, blind conformity has negative consequences, including the transmission of misinformation and the stagnation of innovation, resulting in a lack of progress in cultural evolution (Dean et al., 2014). Methodologically, our results speak to the necessity to consider how different cognitive and social factors affect the generalizability of conformity research with young children. Furthermore, there are important nuances around children's selectivity versus fidelity in copying goals and actions (Over & Carpenter, 2012). These results demonstrate how, within a Western culture, children are willing to deviate from a consensus when it is in their interest in terms of acquiring a larger reward, or when the consensus is perceptually inaccurate and would result in an incorrect response being reported. However, when the cause for a consensus is ambiguous, children conform. Such a finding shows that young children can make informed decisions so as not to simply "follow the crowd." In teaching children, these results suggest that providing clear information about the rationale for a majority's behavior will help children to make incisive decisions across domains about when, and when not, to conform.

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